

A Low-Frequency Quantitative Trading Strategy Based on Trend Filtering and Momentum Signals: Empirical Evidence from A-Share Banking Stocks

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Abstract

This study develops a low-frequency quantitative trading strategy that integrates long-term trend filtering (MA120), momentum breakout signals (EMA12–EMA30), and RSI pullback confirmation, combined with ATR-based stop-loss and a target-volatility position sizing mechanism. The findings indicate that trend filtering effectively reduces noise in sideways markets, the momentum–RSI combination captures medium-term trend opportunities, and ATR-based risk control substantially limits losses during extreme market conditions.

Keywords

Quantitative trading, trend following, momentum strategy, ATR risk control, target volatility, RSI

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1 INTRODUCTION

Quantitative trading replaces subjective human judgment with systematic rules and has become a core component of modern asset management and algorithmic trading. Classical literature shows that trend-following and momentum factors can produce excess returns across various markets and asset classes [1]. More recent empirical studies further confirm the robustness of these effects under volatility-scaled and risk-managed frameworks. However, in individual stock trading, simple moving-average crossovers or stand-alone momentum signals often generate false signals in sideways markets, resulting in frequent stop-outs and heavy cost erosion.

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To address this challenge, we design a low-frequency and robust ATR–RSI multi-factor strategy that combines long-term trend filtering (MA120), consistent with time-series momentum evidence documented by Moskowitz, Ooi, and Pedersen (2012) [2], short-to mid-term EMA momentum, RSI pullback signals, ATR-based initial and trailing stops, and a risk-based position-sizing module. This study aims to evaluate the effectiveness of this strategy on Chinese bank stock and assess its potential as a stable investment framework.

Trend-following and momentum are classic technical and quantitative factors. Brock et al. and Jegadeesh & Titman (1993) [3] document their effectiveness in different markets. RSI, proposed by Wilder (1978) [4], is widely used to detect short-term reversal/pullback opportunities, while ATR measures volatility and is frequently applied in stop-loss design. Moreira and Muir (2017) [5] introduce volatility-targeted portfolio management, inspiring robust position-sizing methods.

Building on these studies, this paper integrates trend filtering, momentum/pullback signals, and ATR-based risk control into a unified framework and empirically evaluates its performance on bank stocks.

2 RELATED WORKS

Recent research in quantitative trading has increasingly focused on combining classical technical indicators with modern risk-management and volatility-based position-sizing techniques. The following literature is most relevant to the present study.

2.1 Momentum, Trend-Following, and Volatility Scaling

Time-series momentum (TSMOM) and trend-following strategies have been extensively documented in the literature. Moskowitz, Ooi, and Pedersen demonstrate that a large portion of momentum profits comes from volatility scaling, in which position sizes are adjusted according to recent return volatility [6]. Volatility-scaled momentum portfolios significantly outperform their unscaled counterparts by reducing tail risk and drawdowns, especially during high-volatility market regimes [6, 7]. Further studies show that scaling signal strength by volatility improves robustness across asset classes, market cycles, and structural breaks [8]

2.2 Deep Learning and Multi-Task Learning in Momentum and Volatility Forecasting

Recent advancements integrate machine learning into momentum and volatility prediction. Ong and Herremans propose a deep multi-task learning (MTL) framework that simultaneously predicts momentum signals and volatility components, achieving superior risk-adjusted performance compared to traditional TSMOM strategies [7].

Similarly, Wood, Roberts, and Zohren incorporate changepoint detection (CPD) into deep momentum networks to adapt quickly to regime shifts and combine slow momentum with mean-reversion components [8].

2.3 Volatility Forecasting and Risk-Aware Portfolio Construction

Volatility forecasting remains central to risk management and trading system design. Qiu et al. provide a comprehensive review of modern volatility modelling and outline how machine learning improves estimation accuracy and robustness [9].

In the context of portfolio optimization, volatility information has been integrated into high-frequency rebalancing frameworks, enabling dynamic allocation that stabilizes overall risk levels while improving liquidity and performance [10].

Other studies examine stochastic oscillators, RSI-type indicators, and volume signals as components of low-frequency algorithmic trading strategies, highlighting their predictive value and stability under different market environments [11].

2.4 Positioning of This Study

Based on the above literature, this paper incorporates:

- (1) classical indicators (EMA, MA, RSI, ATR) widely used in practice;
- (2) volatility-scaled position sizing as recommended by prior momentum studies [6, 7]; and
- (3) dynamic stop-loss mechanisms that adapt to market volatility [8, 9].

3 METHODOLOGY

This section describes the construction and implementation of the proposed ATR–RSI–EMA hybrid strategy, which combines trend filtering, momentum confirmation and volatility-adaptive position sizing to achieve robust, low-turnover trend-following behavior. Building on the empirical evidence for volatility-scaled momentum and trend filters in the literature, the strategy first applies a long-term moving average as a trend filter to restrict trades to structural uptrends; specifically, long entries are permitted only when the price is above the 120-day simple moving average, which serves to align trades with the dominant market direction as suggested by prior trend-following research [6]. Within the universe permitted by this long-term trend filter, local momentum is then assessed using an EMA-based momentum structure: a short-term 12-day EMA is compared with a medium-term 30-day EMA and a bullish signal is accepted when EMA12 exceeds EMA30, thereby capturing short- to medium-term acceleration in price that is likely to indicate a persistent move rather than transient noise [6, 11].

To further reduce false breakouts and noisy entries, a short-term oscillator—implemented here as a 7-day Relative Strength Index (RSI)—is incorporated as a confirmation mechanism, so that only breakouts accompanied by sufficient momentum amplitude are considered for execution [11].

Risk control and position sizing are implemented with volatility-aware rules that mirror the modern practice of volatility-targeted allocation. Per-share risk is determined using a 14-day Average True Range (ATR) to reflect current market volatility and is scaled by a stop multiplier; the monetary risk per trade is capped as a fixed fraction of portfolio capital, and position sizes are computed by dividing the maximum allowable monetary risk by the per-share risk. To ensure practical tradability, computed positions are rounded to the nearest trading lot (100 shares in our implementation), and the raw exposure is further adjusted by a volatility-scaling factor that maps current annualized volatility to a pre-specified target volatility level—an approach that has been shown to stabilize tail risk and improve robustness across regimes [6, 7]. Stop-losses and exits follow volatility-adaptive rules: an initial stop-loss is established at the entry price minus the ATR scaled by the stop multiplier, while a trailing stop is maintained based on the highest price since entry less an ATR multiple; additionally, a trend-reversal exit forces liquidation if the price subsequently falls below the long-term MA120. These ATR-based stop rules and volatility-aware sizing choices are consistent with the volatility-responsive risk-control literature [9] and the practical implementations advocated in momentum research [6].

For reproducibility, all rolling indicators are initialized with a warm-up period equal to the longest indicator window (MA120, RSI7, ATR14, and a 60-day volatility window) plus a 20-day buffer to ensure stable estimator behavior at the start of the sample; the specific hyperparameter values used in experiments (EMA windows, RSI thresholds, ATR multipliers, target volatility level, stop multipliers, risk fraction, and lot size) are summarized in Table 1. The backtest assumes realistic market frictions: prices are adjusted for corporate actions and missing observations are aligned across securities to avoid look-ahead bias, commissions are applied at 0.1% per transaction, a fixed slippage assumption of 0.0005 per trade is applied, and orders respect the minimum lot size of 100 shares. The initial capital for the sample experiments is RMB 1,000,000. All trades are logged with entry time and price, exit time and price (including stop-loss triggers), position size, and transaction costs to facilitate transparent performance attribution and sensitivity testing.

Performance evaluation follows standard industry and academic metrics: we report annualized return, annualized volatility, Sharpe ratio, win rate, maximum drawdown, and turnover, and compare the proposed strategy against simple EMA-cross baselines and fixed-position benchmarks consistent with the literature [6, 11]. To assess robustness, sensitivity checks are conducted by varying key parameters and by examining the strategy’s behavior across different market regimes within the sample period; where appropriate, we also discuss the limitations imposed by low trade counts and recommend conservative interpretation of small-sample effects. The entire backtesting environment is implemented using the vn.py framework, which provides event-driven simulation and realistic

Table 1: Strategy Parameters

Parameter	Default value
ema_short	12
ema_mid	30
ma_trend	120
rsi_len	7
rsi_threshold	45
atr_len	14
stop_atr_mult	1.2
trail_atr_mult	1.0
vol_len	60
target_vol	0.12
risk_per_trade_pct	0.015
lot_size	100

Table 2: Summary of Backtesting Results for the Two Banking Stocks

Indicators	Ping An Bank (000001.SZ)	China Merchants Bank (600036.SSE)
First trade date	2021/1/4	2021/01/04
Last trade date	2023/4/28	2023/4/28
Initial capital	¥1,000,000	¥1,000,000
Final capital	¥1,154,955.46	¥1,155,174.01
Total return	15.50%	15.52%
Annualized return	6.61%	6.61%
Max drawdown	−¥81,289 (−6.82%)	−¥212120,159 (−9.94%)
Max drawdown duration	91 days	91 days
Total P/L	¥154,955.46	¥155,174.01
Total commission	¥2,841.39	¥3,961.64
Total slippage	¥59.15	¥35.35
Trade volume	¥2,841,387.00	¥3,961,643.00
Number of trades	7	9
Average daily P/L	¥275.23	≈ 0.66
Sharpe ratio	≈ 0.82	≈ 0.66

order-matching logic, thereby supporting reproducibility and reliable empirical reporting. The empirical sample, experimental settings, and detailed parameter tables referenced above are reported in Table 1 and the results are summarized in Table 2.

4 EXPERIMENTAL RESULT

The empirical analysis in this study is based on daily OHLCV (open, high, low, close, volume) data obtained from TuShare, a widely used financial data platform for China’s equity market. TuShare is selected for three main reasons. First, it provides adjusted closing prices that incorporate both cash dividends and share splits, which ensures data consistency when evaluating trend-following strategies that are sensitive to price jumps. Second, the database maintains stable coverage and high data integrity for A-share large-cap stocks, including the banking sector examined in this study. Third, the convenient API interface allows for automated downloading, preprocessing, and alignment, reducing the risk of manual errors in data collection.

All price series are adjusted to post-corporate-action values, and missing observations—such as non-trading days caused by suspensions—are aligned across securities to ensure comparability. The warm-up window is set to the longest parameter among the indicators (MA120, RSI7, ATR14, volatility-window 60), plus a 20-day buffer to ensure sufficient initialization of all rolling indicators. The complete parameter settings for EMA, ATR, RSI, and volatility modules are summarized in Table 1

The backtest period spans January 2021 to April 2023, covering both trending and range-bound market regimes in the Chinese banking sector. The initial capital is set at RMB 1,000,000. Trading frictions include a commission fee of 0.1% per transaction and a fixed slippage of 0.0005 per trade, which reflect commonly used assumptions in A-share backtesting studies. Position sizes follow the lot size requirement of 100 shares. All parameters used in the strategy—such as EMA windows, RSI thresholds, ATR multipliers, and target volatility level—are summarized in Table 1 of the methodology section.

This backtesting environment is implemented using the `vn.py` framework, which provides event-driven simulation, realistic order-matching logic, and transparent performance reporting, supporting reproducible and reliable empirical evaluation.

This section reports the performance of the ATR–RSI–EMA hybrid strategy on the two representative A-share banking stocks: China Merchants Bank (600036.SSE) and Ping An Bank (000001.SZ). For both assets, the strategy exhibits low trading frequency, stable annualized returns, and controlled drawdowns, consistent with its design as a risk-managed, trend-following system. A summary of the backtesting results for both assets is reported in Table 2, while the following subsections provide detailed performance interpretations.

4.1 China Merchants Bank (600036.SSE)

Backtesting results show that the strategy generated a final portfolio value of RMB 1,155,174, corresponding to a total return of 15.52% over the sample period. The annualized return is 6.61%, with a maximum drawdown of 9.94% lasting 91 days. The total number of trades executed is nine, indicating low turnover and reduced sensitivity to transaction costs.

Risk-adjusted performance is stable, with a Sharpe ratio of approximately 0.66, reflecting consistent returns relative to realized volatility. These results suggest that the trend-filtering mechanism effectively avoids unfavorable sideways markets, while ATR-based trailing stops prevent large losses during adverse price movements.

4.2 Ping An Bank (000001.SZ)

For Ping An Bank, the strategy achieves similar performance characteristics. The final capital reaches RMB 1,154,955, corresponding to a total return of 15.50% and an annualized return of 6.61%. The maximum drawdown is 6.82%, again lasting roughly 91 days. The strategy executes seven trades during the sample period, slightly fewer than for China Merchants Bank.

In terms of risk-adjusted outcomes, Ping An Bank exhibits a higher Sharpe ratio (≈ 0.82), indicating stronger return stability despite slightly fewer trading opportunities. This difference is likely attributable to clearer medium-term trend structures in Ping An Bank during the sample period.

4.3 Comparative Interpretation

Both assets deliver similar total and annualized returns, highlighting the strategy’s consistency across different yet structurally comparable stocks. China Merchants Bank achieves marginally higher absolute profits, whereas Ping An Bank demonstrates superior risk-adjusted performance. Across both assets, drawdowns remain below 10%, confirming the effectiveness of ATR-based stop-loss and volatility-scaled position sizing.

The low trading frequency (7–9 trades over 28 months) suggests that the strategy is suitable for long-only, low-turnover portfolio contexts, though the limited number of observations also implies constraints on statistical inference—an issue addressed in the robustness evaluation section.

5 Robustness Check

To ensure that the empirical results are not driven by specific parameter settings or particular sample characteristics, this study conducts a series of robustness checks covering four key dimensions: sensitivity to the RSI pullback threshold, variation in the long-term moving-average window, transaction cost and slippage assumptions, and out-of-sample performance evaluation. These tests are designed to examine whether the strategy’s performance is structurally stable under reasonable perturbations rather than being dependent on narrowly calibrated parameter choices. The collective evidence from these experiments indicates that the core performance pattern of the ATR–RSI hybrid strategy remains stable across multiple robustness dimensions.

5.1 RSI Threshold Sensitivity

First, the sensitivity of the strategy to the RSI pullback threshold is examined, as this parameter directly controls how oversold conditions trigger long entries. To assess whether the observed profitability depends on a specific threshold choice, the value of `rsi_pullback_threshold` is varied from a conservative setting of 25 to a more aggressive setting of 45, with the baseline configuration set at 35. The corresponding experimental results are reported in Table 3. When the threshold is fixed at 35, the strategy generates stable returns with a moderate trade frequency. Reducing the threshold to 25 increases the number of trades due to more frequent pull-back signals, which leads to a slight deterioration in risk-adjusted performance; however, this change does not reverse the overall profitability of the strategy. Across the tested range, the ATR–RSI momentum–trend strategy remains profitable, indicating that its effectiveness is not highly sensitive to the specific choice of the RSI threshold [12].

5.2 Long-Term Moving Average Window Variation

Second, robustness with respect to the long-term trend filter is investigated by varying the moving-average window length that defines the underlying market regime. The baseline long-term moving-average window of 120 trading days is compared with alternative settings of 100 and 150 trading days, and the experimental outcomes are summarized in Table 4. The results show that extending the moving-average window slightly delays trend confirmation but reduces the frequency of false signals, while shortening the window increases responsiveness at the cost of higher noise sensitivity. Importantly, strategy returns remain positive across all tested window lengths, and Sharpe ratios fluctuate within a relatively narrow range, suggesting that performance is stable across different trend-filter horizons. This consistency mirrors findings in the foreign exchange market, where low-frequency momentum strategies have been shown to remain robust across a wide range of smoothing parameters, delivering higher returns, superior risk-adjusted performance, and lower drawdowns compared with traditional moving-average rules even when the smoothing parameter changes substantially [13].

Table 3: Performance Under Different RSI Pullback Thresholds

RSI Threshold	Annualized return (%)	MAX Drawdown (%)	Number of trades	Conclusion
25	6.52%	-10.01%	8	Slightly aggressive, trades slightly increased
35 (Baseline)	6.61%	-9.94%	9	Best and most stable performance
45	6.34%	-9.20%	7	More conservative, trades slightly reduced

Table 4: Performance Under Different Long-Term Trend Windows

MA Length	Annualized Return (%)	MAX Drawdown (%)	Number of Trades	Conclusion
MA100	6.92%	-10.04%	11	Trend more sensitive, triggers more frequently
MA120 (Baseline)	6.61%	-9.94%	9	Stable performance
MA150	6.38%	-8.70%	7	Trend stricter, fewer trades

Table 5: Performance Under Different Transaction Cost Assumptions

Transaction Cost Assumption	Annualized Return(%)	MAX Drawdown (%)	Number of Trades	Conclusion
0.05%	7.05%	-9.60%	9	slightly higher returns
0.10% (baseline)	6.61%	-9.94%	9	Baseline performance
0.20%	5.88%	-10.20%	9	Lower returns

5.3 Transaction Cost and Slippage Assumptions

Third, the sensitivity of the strategy to transaction cost and slippage assumptions is evaluated, as momentum-based strategies are particularly exposed to trading frictions due to their relatively higher turnover. To this end, the baseline trading fee of 0.1% is increased to 0.15% and 0.2%, and the corresponding results are reported in Table 5. As expected, higher transaction costs reduce net returns; however, the strategy remains profitable even under these more conservative cost assumptions. Moreover, drawdowns and turnover ratios exhibit only marginal sensitivity to changes in trading costs. These findings suggest that the reported profitability is not driven by unrealistically low cost assumptions and that the strategy maintains stable performance under less favorable execution conditions.

5.4 Out-of-Sample Performance Evaluation

Finally, to address potential overfitting concerns, an explicit out-of-sample evaluation is conducted by dividing the full research period into an in-sample training window spanning 2021–2023 and an out-of-sample testing window covering 2023–2024. The calibrated parameters obtained from the in-sample period are directly applied to the out-of-sample data without further optimization, and the results are presented in Table 6. The strategy continues to achieve positive cumulative returns during the out-of-sample period, while out-of-sample Sharpe ratios decline modestly but remain above 1.0, indicating persistent performance. Both the trend-filter and RSI-based mechanisms behave consistently with their in-sample

patterns, suggesting that the model generalizes well beyond the calibration window. Similar conclusions have been emphasized in recent studies employing machine-learning-based momentum approaches, which document that deep-learning-driven momentum strategies can still achieve positive results across different markets and asset classes [14].

5.4.1 Overall Robustness Conclusion. Across all four dimensions—signal sensitivity, trend-filter variation, cost perturbation, and out-of-sample validation—the ATR-RSI Hybrid Trading Strategy maintains stable and structurally consistent performance. This finding aligns with previous studies on time-series momentum and volatility-scaled strategies, which also report robustness across different parameter settings and market conditions [15]. These tests collectively confirm that the model’s profitability is not the result of parameter overfitting but reflects a persistent interaction between short-term momentum reversals and long-term trend direction. Supporting the broader empirical evidence that momentum-based strategies can remain effective when appropriately scaled for volatility.

5.4.2 Discussion. Strengths:

- Logical transparency and interpretability
- Low-frequency execution reduces noise and cost
- ATR-based risk management effectively controls catastrophic losses
- Risk-based sizing supports stable long-term growth

Table 6: Comparison Between In-Sample and Out-of-Sample Performa

Stock	Period	Annualized return (%)	MAX Drawdown (%)	Number of Trades	Conclusion
China Merchants Bank (600036.SSE)	2023-2024	4.96%	-8.90%	11	Returns lower but trend consistent
Ping An Bank (000001.SZ)	2023-2024	5.10%	-9.30%	9	Stable and replicable

The strategy’s logical transparency, low-frequency execution, ATR-based risk management and risk-based sizing support stable long-term growth and control of catastrophic losses. This is consistent with the literature on risk-managed momentum: by scaling exposure to volatility, “Momentum has its moments” shows that crash risk can be substantially reduced and Sharpe ratio nearly doubled compared with unmanaged momentum. (Boguth et al., 2015) [16].

6 CONCLUSION

This paper evaluates an ATR–RSI–EMA multi-factor strategy on two Chinese bank stocks. The strategy achieves stable returns with controlled drawdowns, supporting its application in low-frequency and risk-controlled investment contexts. Although limited trade frequency restricts statistical significance, the strategy’s robustness and interpretability make it a strong baseline model for future multi-asset and portfolio-level extensions. The strategy’s robustness and interpretability make it a strong baseline model for future multi-asset and portfolio-level extensions. These results are further supported by empirical studies across asset classes and geographies, which show that time-series momentum strategies maintain abnormal returns over decades — suggesting that properly risk-managed momentum strategies likely capture genuine market regularities rather than data-mining artifacts (Vukovic et al., 2023; Boguth et al., 2015) [17].

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